

Tools for Security Audit and Vulnerability Analysis of Smart Contracts

Smart contracts are an essential component of blockchain technology, and have helped it transcend typical cryptocurrency and financial transactions. But these contracts do come with a few flaws, making it imperative that they go through a security audit. Among the various tools available for this, Slither ranks high.

Blockchain technology has emerged as a revolutionary solution with versatile applications beyond its initial association with cryptocurrencies. Its decentralised and tamper-resistant nature offers a range of real-world use cases that go beyond financial transactions.

With the implementation of smart contracts, the blockchain environment provides high-end and fully secured applications for assorted domains including e-governance, real estate,

finance, logistics, and many others. Smart contracts are transforming the real estate industry by automating property transactions. They facilitate seamless and secure buying, selling, and renting of properties, with ownership transfers occurring only when payment conditions are met.

Key use cases and applications of blockchain

Imagine a world where transparency is a given, fraud is rendered nearly impossible, and trust between parties is inherent. That's the world blockchain technology is shaping.

- **Supply chain management:** Blockchain provides end-to-end transparency and traceability in supply chains. Each step of a product's journey can be recorded on the blockchain, ensuring authenticity and preventing fraud. This is particularly important in industries where provenance, such as the origin of luxury goods or food products, is critical. Smart contracts ensure automatic execution of agreements when predefined conditions, such as the arrival of goods at a specified location, are met. This real-time tracking minimises fraud, reduces paperwork, and streamlines logistics processes.
- **Decentralised finance (DeFi):** Smart contracts have become the foundation of the DeFi movement. They power lending and borrowing platforms, decentralised exchanges (DEXs), automated market makers (AMMs), and yield farming

